

Solution Requirements

Date	03 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The system should allow authorized users to securely access the application through a login mechanism.</i>	Medium
2	Authorization levels	<i>The system should provide appropriate access permissions. Customers can submit applicant details and view predictions, while administrators can manage the deployed model and monitor system usage.</i>	Medium

3	External interfaces	<i>The system should integrate with IBM Watson Machine Learning for cloud-based model deployment and prediction services and provide a web interface developed using Flask.</i>	High
4	Transactions processing	<i>The system should accept applicant information, preprocess the input data, execute the machine learning model, and return an approval or rejection prediction in real time.</i>	High
5	Reporting	<i>The system should display the prediction result to the user and provide model performance metrics such as accuracy, precision, recall, and F1-score during model evaluation.</i>	Medium
6	Business rules, etc.	<i>The prediction should be generated based on applicant financial and demographic information using the trained machine learning model. Applications with higher financial risk should be classified as less likely to receive approval.</i>	High
7	Compliance to laws or regulations	<i>The system should maintain applicant data confidentiality and comply with applicable data privacy principles by ensuring that sensitive personal information is handled securely.</i>	High

8	Other	The system should allow users to enter applicant information through an intuitive web interface and receive instant credit card approval predictions.	Medium
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate predictions quickly after the user submits the application details.	Prediction generated within 2 seconds .
2	Scalability	The deployed application should support multiple users simultaneously without significant performance degradation.	Supports 100+ concurrent prediction requests .
3	Security & Data Privacy	Applicant information should be securely processed and protected against unauthorized access.	Secure data transmission using HTTPS and restricted access to sensitive data.
4	Reliability & Availability	The application should remain available for users with minimal downtime.	99% uptime during deployment
5	Usability & Accessibility	The web application should provide a simple, user-friendly interface with clearly labeled input fields and prediction results.	New users can complete a prediction in less than 2 minutes .

6	Other	<i>The system should be easy to maintain, retrain with new datasets, and deploy on different environments such as local systems or cloud platforms.</i>	Model can be updated and redeployed without modifying the application interface.
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